deep se
dependable evolvable pervasive software engineering group

Virtual office hours 1

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POLITECNICO DI MILANO

Recurrent questions

- Pointers
- Why private?
- Complexity
- Polymorphism
- Static

Pointers

- I don’t expect you to decide when to use a pointer or not
  - You need to recognize memory leakage with raw pointers
  - You need to be able to write code once classes structures (with pointers) are given
  - We cover raw pointers part:
    - Because you will encounter code using pointers during your study
    - Because I have to make you aware of the issues behind (otherwise when you are in the situation above you will not be even able to move on and understand where the issues arise, see e.g. assignment operators)

Pointers

- C needs to use pointers for parameters pass-by-reference and to store variable number of items
- If you need to store variable number of items
  - Use STL containers

Pointers

- If you need to create variable number of large objects and you want to avoid copies
  - Use smart pointers
- If you need to access objects stored in the stack (e.g., stored in containers) and you want to avoid copies
  - Use raw pointers
- If you have to manage complex inheritance hierarchies and you need containers, use pointers or smart pointers according to where objects are stored
  - stack -> raw pointers
  - heap -> smart pointers

Why private?

- Preserve an invariant

- Change the internal data structure
  - We will see why this is important next week when introducing the STL containers
  - This links also with complexity

Complexity

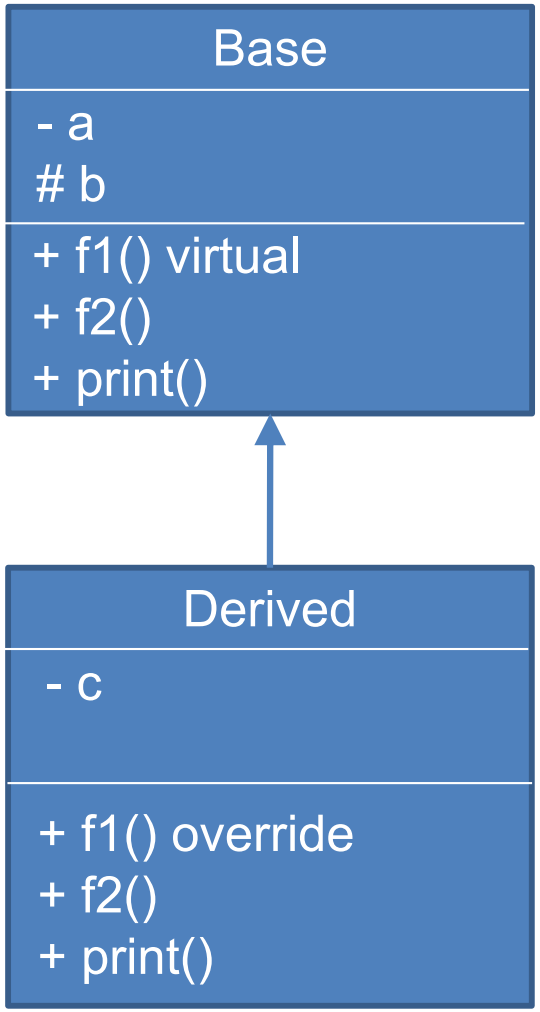
- Let's consider complexity of last lab, Social Network
  - `push_back` in a vector cost O(N), where N is the number of items in the vector

Complexity

- Let's consider complexity of last lab, Social Network
  - push_back in a vector cost O(N), where N is the number of items in the vector

- SocialNetwork::CUserIndex(): O(N)
- SocialNetwork::AddUser(): O(N)
- SocialNetwork::CGetUsers(): O(N)
- SocialNetwork::CGetFriends: O(N^2)
- SocialNetwork::AddFriendship: O(N)

Polymorphism



Static

- You have to compute students taxes for PoliMI

- Student:
  - ID
  - full_name
  - family_income

- Taxes:
  - Low rate: if family_income < α_1 : β_1 * family_income
  - Average rate: if $\alpha_1 \leqq$ family_income < α_2 : β_2 * family_income
  - High rate: if family_income $\geqq \alpha_2$: β_3 * family_income